The battery-only charge hypothesis

1 Problem Setup

The presentation asks whether charge at the battery alone can make an electric field that guides current around a bent circuit. For this comparison, assume that the two battery terminals carry uniform, opposite charge densities. Put no charge on the wire surfaces or resistor interfaces. This is deliberately not the conducting circuit's solution.

As in the rest of the app, the geometry is constant in the z direction. A terminal edge in the xy plane extends indefinitely along z. We calculate its field at the center of every triangle of the unified mesh.

2 Finding the field of a charged terminal edge

Start with a line charge of strength λ per unit z length. A circular Gaussian surface of radius r and unit depth has outward electric flux $E(2\pi r)$. Gauss's law therefore gives $E = \lambda/(2\pi\epsilon r)$. In vector form, the field at $\boldsymbol{r}$ from charge located at $\boldsymbol{r}'$ is proportional to $(\boldsymbol{r} - \boldsymbol{r}')/|\boldsymbol{r} - \boldsymbol{r}'|^2$.

For a terminal mesh edge e with constant charge density σ_e, a small piece ds' contributes line charge $\sigma_e\, ds'$. Add all pieces of the edge:

$$\boldsymbol{E}_e(\boldsymbol{r}) = \frac{\sigma_e}{2\pi\epsilon} \int_e \frac{\boldsymbol{r} - \boldsymbol{r}'}{|\boldsymbol{r} - \boldsymbol{r}'|^2}\, ds'. \tag{1}$$

The total toy field is the sum of equation (1) over both terminals. The app evaluates each edge integral numerically with 12-point Gauss–Legendre quadrature. The evaluation points are triangle centers, away from the charged edges. This field spreads into the air and becomes weaker farther away; it does not respond to the shape of the rest of the wire.

3 Choosing the amount of terminal charge

The toy hypothesis does not tell us the charge density. We choose one scale so that its field inside the battery is comparable with the actual battery field. Let the battery EMF be $\mathcal{E}$, its length ℓ_b, and its driving direction $\boldsymbol{d}_b$. The conduction law inside the battery is

$$\boldsymbol{J} = \frac{\boldsymbol{E} + \boldsymbol{F}}{\rho_b}, \qquad \boldsymbol{F} = \frac{\mathcal{E}}{\ell_b}\boldsymbol{d}_b. \tag{2}$$

In the limit of very small battery resistivity ρ_b and finite current, $\boldsymbol{E}$ approaches $-\boldsymbol{F}$. We use the axial value $-\mathcal{E}/\ell_b$ as the calibration target.

First give one terminal density $+1$ and the other -1, and calculate the resulting field $\boldsymbol{E}^{(1)}$. For the battery triangles B, compute its area-weighted response along the battery direction:

$$R = \frac{\sum_{h\in B} A_h\, \boldsymbol{E}_h^{(1)} \cdot \boldsymbol{d}_b}{\sum_{h\in B} A_h}. \tag{3}$$

Electric field is linear in charge density, so multiplying both terminal densities and the trial field by

$$\alpha = \frac{-\mathcal{E}/\ell_b}{R} \tag{4}$$

makes the battery-average axial field match the target. The app displays $\boldsymbol{E}_{\text{toy}} = \alpha \boldsymbol{E}^{(1)}$ and the corresponding terminal charges. This calibration does not make the toy field satisfy the wire's boundary conditions.

Code: `calculations/battery_only.py`.